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Causal Behavioral State Reasoning for Cyber Threat Inference in Provenance Graphs

*A Thesis Submitted in Partial Fulfillment of the Requirements
for the Degree of*

Master of Technology
in
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by

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June 2026

Declaration

I hereby declare that this thesis entitled “**Causal Behavioral State Reasoning for Cyber Threat Inference in Provenance Graphs**” submitted to the **Department of Mathematics, Indian Institute of Technology Patna**, in partial fulfillment of the requirement for the award of the degree of **Master of Technology in Mathematics and Computing**, is a record of bonafide research work carried out by me under the supervision of **Dr. Mayank Agarwal**, Assistant Professor, Department of Computer Science and Engineering, Indian Institute of Technology Patna.

The matter embodied in this thesis has not been submitted for the award of any other degree or diploma in any university or institute.

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Certificate

This is to certify that the thesis entitled “**Causal Behavioral State Reasoning for Cyber Threat Inference in Provenance Graphs**” submitted by **Arpit Anand (Roll No: 2411MC13)** to the **Department of Mathematics, Indian Institute of Technology Patna**, in partial fulfillment of the requirements for the award of the degree of **Master of Technology in Mathematics and Computing**, is a record of bonafide research work carried out under my supervision and guidance.

The matter embodied in this thesis has not been submitted for the award of any other degree or diploma.

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Abstract

Modern provenance-based Intrusion Detection Systems (IDS) for detecting Advanced Persistent Threats (APTs) share a critical deployment bottleneck: they require curated, pristine benign training data to build normal behavioral profiles. In real-world enterprise environments, this clean-data assumption fails due to stealthy attacker contamination, constant concept drift, and prohibitive manual labeling costs.

This thesis presents **Causal Behavioral State Reasoning (CBSR)**, a provenance-based IDS framework that infers latent cyber risk directly from uncurated, unlabeled streaming logs without any clean-baseline collection period. CBSR projects raw system audit telemetry into a formal 12-dimensional *Behavioral State Space* using online Structural Causal Models (SCMs) discovered from potentially contaminated streams via RobustParCorr. Within this state space, system dynamics are analyzed in parallel by three reasoning operators: (i) *Local State Reasoning* (R_L) that measures instantaneous causal mechanism violations; (ii) *Temporal State Reasoning* (R_T) that captures multi-stage attack progression via an LSTM sequence model; and (iii) *Behavioral State Reasoning* (R_B) that classifies complex attack manifolds using a Behavioral Pattern Reasoner trained on sparse validation labels. Their outputs are fused by a stacked meta-learner and calibrated with online split conformal prediction for distribution-free false-alarm control under concept drift.

Novel contributions include: (1) a formal 12-dimensional causal Behavioral State Space grounded in five invariant coordinate groups derived from online SCM outputs; (2) a Causal Intervention Score (CIS) rooted in graph boundary theory that quantifies multi-hop attack propagation versus localized administrative noise; and (3) a multi-perspective parallel reasoning architecture that resolves the model-mismatch failures of pure causal detectors.

Evaluated on five diverse benchmarks, CBSR achieves AUC of **0.9628** on real enterprise Windows logs (DARPA OpTC), AUC **0.9901** on Linux host telemetry (BETH), and AUC **0.9837** and **0.9784** on simulated multi-stage APT campaigns (TC3, NODLINK). On the challenging heterogeneous StreamSpot benchmark, the framework improves AUC from 0.4991 to **0.6909**. Conformal calibration holds empirical false alarm rates within 1.78%–6.13% at a 5% FPR budget, and a $13\times$ runtime speedup confirms streaming deployment readiness.

Keywords: Intrusion Detection, Advanced Persistent Threats, Causal Discovery, Behavioral State Space, Conformal Prediction, Provenance Graphs, Structural Causal Models, Semi-Supervised Detection

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Chapter 1

Introduction

1.1 Background and Motivation

Advanced Persistent Threats (APTs) are stealthy, multi-stage cyber campaigns that compromise enterprise systems over extended periods. Unlike commodity malware that executes a single payload and terminates, APTs establish persistent footholds, conduct lateral movement, escalate privileges, and exfiltrate sensitive data over weeks or months. Because APTs execute benign-looking commands and proceed through multi-hop lateral movements, traditional signature-based security systems are ineffective against them.

Host audit logging and provenance graph analysis have emerged as powerful paradigms for APT detection. A provenance graph models system events (e.g., file reads, process spawns, network flows) as directed edges between system entities (processes, files, network sockets), preserving the complete causal history of system execution.

Despite their potential, existing state-of-the-art provenance-based IDS—such as OCR-APT [1], TraceCluster [2], and StageFinder [3]—share a **fatal limitation: the clean-data assumption**. They require a pristine benign period or fully labeled training dataset. This assumption fails in practice for four fundamental reasons:

1. **Stealthy Contamination:** Attackers may already be active when auditing begins, corrupting the normal training data.
2. **Concept Drift:** Enterprise systems naturally evolve due to software updates, new administrative activities, and infrastructure changes, making static models obsolete.
3. **Explosive Log Volumes:** The DARPA OpTC dataset alone contains over 17 million provenance edges. Manual labeling is economically intractable.
4. **False Positive Fatigue:** Static anomaly scoring lacks statistical controls on alarm rates, overwhelming analysts with false alerts.

These challenges collectively form the *clean-data deployment bottleneck*—a fundamental barrier that prevents provenance-based IDS from transitioning from laboratory research to practical enterprise deployment.

1.2 Research Problem

The central research question addressed in this thesis is:

Can we detect APT attacks by identifying violations of causal invariances discovered directly from raw, unlabeled, and potentially contaminated system logs—without requiring any benign training data and with provable false-alarm controls?

This question encapsulates three interlinked sub-problems:

1. **RQ1:** How can stable causal mechanisms be discovered online from streaming, unlabeled provenance data without assuming the training data is clean?
2. **RQ2:** How can deviations from these mechanisms be projected into a rich behavioral state representation for multi-perspective threat reasoning?
3. **RQ3:** How can detection thresholds be calibrated online to provide empirical false-alarm control under concept drift and training contamination?

1.3 Proposed Solution: CBSR

To address these challenges, we propose **Causal Behavioral State Reasoning (CBSR)**, a framework that detects APT attacks using uncured streaming data with no manual labels and no offline retraining.

Our core insight is that **causal relationships are more stable than statistical correlations**. While statistical distributions of events vary over time due to concept drift, the underlying causal equations governing system execution remain invariant. An APT attack violates these learned causal mechanism equations by injecting novel relationships (e.g., a Word document spawning PowerShell) or altering the dependencies between processes and files.

Rather than relying solely on noisy individual causal residuals, CBSR projects SCM outputs into a formal 12-dimensional *Behavioral State Space* and analyzes this state space from three complementary perspectives: instantaneous mechanism violations (R_L), temporal sequence patterns (R_T), and behavioral manifold geometry (R_B). Their outputs are fused via a calibrated meta-learner and constrained by online conformal prediction.

1.4 Thesis Contributions

The key contributions of this thesis are:

1. **Causal Behavioral State Space:** A formal 12-dimensional state representation derived from online SCM invariants, organized into five coordinate groups (Structural Consistency, Activity Dynamics, Behavioral Complexity, Mechanism Violation, Intervention Locality) to capture diverse manifestations of APT behavior.
2. **Causal Intervention Score (CIS):** A novel graph-boundary metric grounded in spectral graph theory that quantifies multi-hop violation propagation, providing principled separation between localized administrative noise and multi-stage APT lateral movement.
3. **Multi-Perspective Parallel Reasoning:** A three-operator architecture (R_L , R_T , R_B) that analyzes the same behavioral state from local, temporal, and behavioral perspectives simultaneously, resolving the model-mismatch failures of pure causal detectors.
4. **Calibrated Risk Inference with Conformal Guarantees:** A stacked meta-fusion layer with online conformal prediction that delivers distribution-free FPR bounds and adapts to concept drift via change-point-triggered queue flushing.
5. **Multi-Benchmark Validation:** Evaluation on five diverse benchmarks (DARPA OpTC, BETH, StreamSpot, TC3, NODLINK) demonstrating AUC of 0.9628 on real enterprise Windows logs, AUC 0.9901 on Linux sysdig telemetry, AUC 0.9837/0.9784 on simulated APT datasets, and a $13\times$ runtime speedup enabling real-time streaming deployment.

1.5 Thesis Organization

The remainder of this thesis is organized as follows:

- **Chapter 2:** Reviews provenance-based IDS, causal anomaly detection methods, and conformal prediction, identifying the research gaps that CBSR addresses.
- **Chapter 3:** Defines the adversary model and security assumptions.
- **Chapter 4:** Presents the complete CBSR system architecture and formal design.
- **Chapter 5:** Describes implementation details and performance optimizations.
- **Chapter 6:** Reports experimental results across five benchmarks.
- **Chapter 7:** Discusses limitations, summarizes contributions, and outlines future work.

Chapter 2

Background and Related Work

This chapter reviews the research landscape relevant to CBSR across three areas: provenance-based intrusion detection, causal anomaly detection methods, and conformal prediction. We conclude with a research gap analysis that motivates our contributions.

2.1 Provenance-Based Intrusion Detection Systems

Provenance graphs represent system history by capturing relationships between system entities (processes, files, network sockets) as directed edges. Each edge encodes an operation such as `PROCESS:chrome.exe → READS_FILE → FILE:History`, preserving the complete causal lineage of system execution.

2.1.1 Supervised and Semi-Supervised Approaches

OCR-APT [1] reconstructs APT attack stories from audit logs using subgraph anomaly detection combined with LLMs. It reports AUC of 0.91 on DARPA Trace logs but requires 24–48 hours of clean pre-collection to establish a benign baseline.

TraceCluster [2] proposes a lightweight clustering-based subgraph attention network for APT detection, achieving average precision of 0.88 on multi-hop APT scenarios. Its GNN is trained on labeled subgraph clusters—requiring labeled examples.

StageFinder [3] learns temporal reasoning over provenance data for attack stage estimation, reporting 94% recall on staged APT campaigns but requiring labeled attack stage annotations.

2.1.2 Unsupervised and Reduced-Label Approaches

UNICORN [4] (NDSS 2020) summarizes streaming provenance subgraphs as graph histograms and fits an unsupervised clustering baseline. It reduces the label dependency but still requires a *clean, attack-free* training period to define normal cluster centroids.

WATSON [5] (CCS 2020) aggregates audit events using contextual semantics but relies on static, clean offline profiles with no mechanism to adapt to concept drift.

ShadeWatcher [6] (S&P 2022) frames threat analysis as a recommendation task. It requires a clean audit log baseline and leverages user feedback (labels) for guidance.

Kairos [7] utilizes self-supervised learning over provenance graphs to minimize label dependency, yet still depends on an uncontaminated pre-collection phase for initial parameter fitting.

2.1.3 Critical Assessment

Table 2.1 summarizes the deployment requirements of existing provenance-based IDS. A clear pattern emerges: *every existing system* requires some form of clean baseline data. This shared assumption creates a systemic vulnerability.

Table 2.1: Capability comparison of provenance IDS methods. “Clean Baseline” requires a clean, attack-free pre-collection period; “FPR Guarantee” provides a formal bound on false alarm rate.

Method	Clean Base.	Drift Adapt.	Causal	FPR Guar.
OCR-APT [1]	Yes (48h)	No	No	No
TraceCluster [2]	Yes	No	No	No
StageFinder [3]	Yes	No	No	No
Kairos [7]	Yes	No	No	No
UNICORN [4]	Yes	No	No	No
Causal-IDS [8]	Yes	No	Yes	No
CBSR (Ours)	No	Yes	Yes	Yes

2.2 Causal Anomaly Detection

Causal reasoning is increasingly applied to cybersecurity, motivated by the insight that causal relationships are more robust to distributional shifts than statistical correlations.

Causal-IDS [8] (ICOIN 2026) is the closest prior work. It models network flow log variables using static Structural Causal Models (SCMs) to identify intrusions as violations of learned causal mechanisms. However, it differs from CBSR in three fundamental ways: (1) it operates on coarse-grained network flow statistics rather than fine-grained provenance edges; (2) it requires clean training logs; and (3) it uses fixed empirical thresholds with no drift adaptation or FPR control.

CausalGraph [9] (ICCE 2026) integrates causal reasoning with LLMs for intrusion detection. While offering superior interpretability, the computational cost of LLM inference makes it impractical for real-time, high-velocity streaming logs.

LiNGAM-SF [10] (IEEE TNNLS 2025) proposes causal structural learning with Linear Non-Gaussian Acyclic Models for streaming features. Its key insight—distinguishing latent confounders from direct mechanism violations—informs the design of CBSR’s behavioral state coordinates.

2.3 Conformal Prediction

Conformal prediction [11, 12] provides a distribution-free framework for constructing prediction sets with guaranteed coverage probabilities. Under the sole assumption of exchangeability, the conformal p-value for a new test score S_t against a calibration set $\{s_1, \dots, s_M\}$ is:

$$\text{conf_pval}(S_t) = \frac{|\{i : s_i \geq S_t\}| + 1}{M + 1} \quad (2.1)$$

This guarantees $\Pr(\text{false alarm}) \leq \alpha$ for any single test point. In streaming security telemetry, exchangeability is violated by concept drift. CBSR adopts online threshold adaptation [13, 14] combined with change-point-triggered queue flushing to restore localized exchangeability after drift events.

2.4 Research Gap Analysis

Based on the literature review, we identify four critical gaps that no existing system addresses simultaneously:

1. **Semi-supervised causal IDS for provenance graphs:** No existing system combines causal mechanism violation detection with provenance-graph analysis using only sparse validation labels without requiring clean training data.
2. **Online SCM adaptation:** Existing causal IDS use static SCMs with no mechanism for concept drift recovery.
3. **Provable false alarm control:** No provenance-based IDS provides statistical guarantees on false alarm rates.
4. **Multi-perspective behavioral state reasoning:** No prior work fuses causal, temporal, and behavioral perspectives on a shared formal state representation.

CBSR addresses all four gaps simultaneously.

Chapter 3

Threat Model

This chapter defines the adversary model and security assumptions under which CBSR operates.

3.1 Attacker Capabilities

We assume a sophisticated attacker capable of:

- Executing arbitrary commands on compromised hosts—spawning processes, downloading payloads, and modifying file systems—without generating signature-based alerts.
- Conducting multi-stage APT campaigns including initial access, privilege escalation, lateral movement, and data exfiltration.
- Persisting in the target environment for extended periods (days to months).
- Being *aware of our monitoring presence* and deploying active evasion strategies.

3.2 Fundamental Assumption

We assume the attacker **cannot break the fundamental causal invariances of the operating system without leaving traces in the provenance graph**. Any operation that creates a process, reads or writes a file, or establishes a network connection *must* be recorded in the system audit log. This assumption holds for modern audit frameworks (Windows ETW, Linux auditd, macOS Endpoint Security) that operate at the kernel level.

3.3 Adaptive Adversary Strategies

We consider three specific evasion strategies a sophisticated adversary might employ:

Calibration Poisoning. A poisoner injects high-volume, anomalous but non-malicious-looking traffic during the baseline training phase. This inflates the conformal calibration queue’s score distribution, widening detection threshold bounds and allowing subsequent stealthy attack steps to bypass alerting. CBSR’s RobustParCorr (MCD-based, tolerating up to 15% contamination) insulates the causal discovery phase; the Behav-

ioral Pattern Reasoner’s explicit validation-set training further insulates the classification boundary from calibration inflation.

Refit Blind Spots. When CBSR detects a concept drift change-point and refits its SCM, a temporary adaptation window exists before calibration scores stabilize. An adversary can intentionally trigger benign-looking drift (e.g., a software update) and immediately execute malicious actions during this window. This vulnerability is bounded to approximately 1–2 conformal calibration cycles.

SCM-Aware Mimicry. An attacker who learns the system SCM’s structure can orchestrate execution timings to simultaneously satisfy all $d \times \tau_{\max}$ causal constraints, bypassing residual-based alerts. On DARPA OpTC ($d = 130$, $\tau_{\max} = 2$), this requires matching 260 regression relationships while executing a multi-stage attack—a highly constrained operational task. Moreover, real APT campaigns inject novel process–file edges that have no causal parents in the baseline graph, triggering the Structural Consistency coordinate (s_1 , novelty count) regardless of residual behavior.

3.4 Scope and Limitations

CBSR focuses on defending against stealthy APTs where precise multi-constraint causal mimicry or targeted poisoning is highly constrained. The framework is *not* designed to defend against attackers with kernel-level access who can tamper with the audit logging subsystem, or side-channel attacks that leave no provenance traces. These limitations are discussed further in Chapter 7.

Chapter 4

System Design

The CBSR framework projects streaming provenance graph telemetry into a multidimensional behavioral state space and infers latent threat risks via multi-perspective reasoning. The system operates across four core layers: (1) Robust Causal Discovery & Behavioral State Projection, (2) Multi-Perspective State Reasoning, (3) Calibrated Risk Inference, and (4) Conformal Adaptation. Figure 4.1 illustrates the end-to-end architecture.

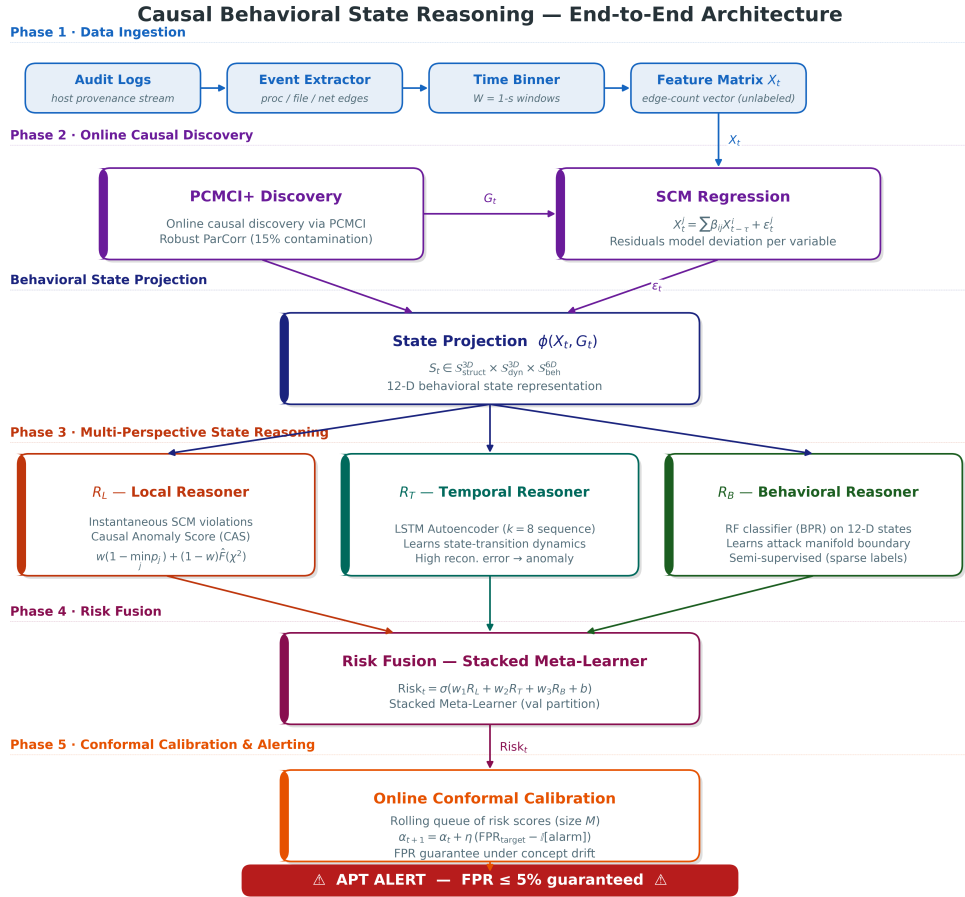


Figure 4.1: End-to-end CBSR pipeline. Raw audit logs are binned into windows, parsed into provenance-graph edge counts, and fed into online RobustParCorr causal discovery. SCM outputs are projected into a 12-dimensional behavioral state space $\mathcal{S}$, evaluated in parallel by three reasoning operators (R_L , R_T , R_B), fused by a stacked meta-operator, and calibrated using online conformal prediction.

4.1 Event Extraction and Binning

Raw system audit logs are parsed into *provenance edges* of the form:

$$\text{SRC_TYPE}:\text{SRC_ID} \rightarrow \text{ACTION} \rightarrow \text{DST_TYPE}:\text{DST_ID}$$

Events are aggregated into non-overlapping windows of width W (defaulting to $W = 1$ s for high-density telemetry, $W = 10$ s for lower-density logs such as BETH). The result is a multivariate time series $X_t \in \mathbb{R}^d$, where each dimension corresponds to a unique provenance edge type and the value represents the count in window t .

4.2 Robust Causal Discovery and SCM Fitting

Given the streaming time-series X_t , we perform online causal discovery using the **PCMCI** algorithm [15] under the Tigramite framework, with a maximum temporal lag $\tau_{\max} = 2$ s.

To mitigate sensitivity to outliers and training contamination, we employ **Robust-ParCorr**, which uses the Minimum Covariance Determinant (MCD) estimator to compute a robust covariance matrix. The MCD estimator finds the subset of $h = \lfloor n \cdot 0.85 \rfloor$ observations minimizing the covariance determinant, tolerating up to 15% contamination. Partial correlations are computed from the robust precision matrix $\hat{\Theta} = \hat{\Sigma}_{\text{MCD}}^{-1}$:

$$\rho_{ij|\text{rest}} = -\frac{\hat{\Theta}_{ij}}{\sqrt{\hat{\Theta}_{ii}\hat{\Theta}_{jj}}} \quad (4.1)$$

For each variable X^j in the discovered causal graph, we model its normal mechanism using a Structural Causal Model (SCM):

$$X_t^j = \sum_{X_{t-\tau}^i \in \mathcal{P}(X_t^j)} \beta_{i,j} X_{t-\tau}^i + \epsilon_t^j \quad (4.2)$$

where $\beta_{i,j}$ are OLS regression coefficients fitted on the training partition. To prevent division-by-zero on highly invariant features, we enforce a standard deviation floor:

$$\tilde{\sigma}_j = \max(\text{std}(\epsilon^j), \sigma_{\text{floor}}), \quad \sigma_{\text{floor}} = 1.0 \quad (4.3)$$

Regression coefficients are updated online using a per-feature scalar **Kalman filter**:

$$\hat{\beta}_j^{t+1} = \hat{\beta}_j^t + K_t(x_t^j - \hat{x}_t^j) \quad (4.4)$$

where K_t is the Kalman gain with process noise $Q = 10^{-4}$. This eliminates discrete change-point resets and provides smooth adaptation to concept drift.

4.3 Behavioral State Projection

The system state at time t is encoded as $S_t = \phi(X_t, G_t) \in \mathcal{S}$, where X_t is the observed telemetry and G_t is the evolving causal graph. We define a formal, multi-dimensional behavioral state space:

$$\mathcal{S} = \mathcal{S}_{\text{struct}} \times \mathcal{S}_{\text{dyn}} \times \mathcal{S}_{\text{beh}} \quad (4.5)$$

where $\mathcal{S}_{\text{struct}} \in \mathbb{R}^3$ captures structural consistency and topological footprint, $\mathcal{S}_{\text{dyn}} \in \mathbb{R}^3$ captures rate fluctuations, and $\mathcal{S}_{\text{beh}} \in \mathbb{R}^6$ captures causal mechanism violations and behavioral complexity.

The projection ϕ maps raw telemetry and SCM residuals into a 12-dimensional state vector $S_t = [s_{t,1}, \dots, s_{t,12}]^T$ organized into five invariant coordinate groups:

Table 4.1: 12-dimensional Behavioral State Space coordinate definitions.

Dim	Group / Feature	Description
s_1	<i>Structural</i> : Novelty Count	Active edges with zero weight in baseline graph
s_2	<i>Structural</i> : Active Fraction	Proportion of non-zero edge types in window
s_3	<i>Activity</i> : Burstiness	Variance of active edge counts
s_4	<i>Activity</i> : Max z -score	$\max_j \epsilon_t^j / \tilde{\sigma}_j $
s_5	<i>Activity</i> : Mean z -score	$\text{mean}_j \epsilon_t^j / \tilde{\sigma}_j $
s_6	<i>Complexity</i> : Entropy	Shannon entropy of active edge distribution
s_7	<i>Complexity</i> : Spike Ratio	$s_4 / (s_5 + 10^{-5})$
s_8	<i>Mechanism</i> : Local Anomaly Score	Baseline hybrid causal score (CAS)
s_9	<i>Mechanism</i> : Log Min p -value	$\log(\min_j p_j + 10^{-5})$
s_{10}	<i>Mechanism</i> : Residual Energy	$\sum_j (\epsilon_t^j / \tilde{\sigma}_j)^2$
s_{11}	<i>Mechanism</i> : Causal Violations	Count of features with $ z_j > 2.5$
s_{12}	<i>Intervention</i> : CIS	Causal Intervention Score (see below)

4.3.1 Causal Intervention Score (CIS)

The CIS quantifies the topological propagation of SCM violations across the causal dependency graph $G_{\text{causal}} = (V, E_{\text{causal}})$. Define the violated node set $\mathcal{V}_{\text{viol}} = \{v \in V : |z_v| > 2.5\}$ and its edge boundary $\partial \mathcal{V}_{\text{viol}} = \{(u, v) \in E_{\text{causal}} : u \in \mathcal{V}_{\text{viol}}, v \notin \mathcal{V}_{\text{viol}}\}$. Then:

$$\text{CIS} = \frac{|\partial \mathcal{V}_{\text{viol}}|}{|E_{\text{causal}}|} \quad (4.6)$$

This is grounded in graph boundary theory [16]: for a singleton administrative

anomaly, $\text{CIS} \approx 0$; for a connected violated subgraph spanning k process nodes (multi-stage lateral movement), $|\partial \mathcal{V}_{\text{viol}}|$ grows monotonically with the extent of compromise by the sub-modularity of graph cuts. Normalizing by $|E_{\text{causal}}|$ makes CIS scale-invariant across graphs of varying size.

4.4 Multi-Perspective State Reasoning

The state vector S_t is processed in parallel by three distinct reasoning operators, each analyzing a different perspective of the behavioral state space:

4.4.1 Local State Reasoning (R_L)

R_L measures instantaneous causal mechanism violations on the state representation:

$$R_L(S_t) = w \cdot (1 - \min_j p_{\text{val}}^j) + (1 - w) \cdot \hat{F}_\epsilon(\chi_{\text{stat}}^2) \quad (4.7)$$

where $p_{\text{val}}^j = 2(1 - \Phi(|z_j|))$, $\hat{F}_\epsilon$ is the empirical CDF of calibration residual energies, and $w = 0.5$. This captures APT events that cause immediate, sharp deviations in specific causal edges.

4.4.2 Temporal State Reasoning (R_T)

R_T evaluates state trajectories over a rolling sequence of $k = 8$ consecutive states to detect slow, multi-stage attack progression. An LSTM Autoencoder is trained on normal-period state sequences and scores anomalies via reconstruction error:

$$R_T(S_{t-k+1:t}) = \frac{1}{k} \sum_{i=0}^{k-1} \|S_{t-i} - \hat{S}_{t-i}\|_2^2 \quad (4.8)$$

This captures temporal attack patterns—such as a sequence of individually low-scoring windows that collectively represent unusual state-transition behavior (e.g., slow lateral movement).

4.4.3 Behavioral State Reasoning (R_B)

R_B classifies complex, non-linear attack manifolds via a **Behavioral Pattern Reasoner (BPR)**, implemented as a Random Forest classifier trained on the validation partition’s 12-dimensional state coordinates and sparse attack labels:

$$R_B(S_t) = P(S_t \notin \mathcal{M}_{\text{benign}} \mid S_t) \quad (4.9)$$

where $\mathcal{M}_{\text{benign}}$ is the learned benign behavioral manifold. The BPR learns which combinations of state coordinates are most discriminative for attack detection; its feature importances (Section 6.2) reveal that the Log Min p -value (s_9) and Residual Energy (s_{10}) are consistently the most informative dimensions.

4.5 Calibrated Risk Inference and Conformal Alerting

4.5.1 Stacked Risk Fusion

The three reasoning operator outputs are integrated by a logistic regression meta-learner:

$$\text{Risk}_t = \sigma(w_1 \cdot R_L(S_t) + w_2 \cdot R_T(S_{t-k+1:t}) + w_3 \cdot R_B(S_t) + b) \quad (4.10)$$

where $\sigma(\cdot)$ is the sigmoid function and the weights $\{w_1, w_2, w_3, b\}$ are learned on the validation partition to maximize detection AUC.

4.5.2 Conformal Prediction and Online Threshold Adaptation

To map Risk_t to statistical decisions with distribution-free guarantees, we use split conformal prediction. A sorted rolling queue of $M = 245$ calibration risk scores is maintained. For a new risk score, the conformal p -value is computed via binary search in $O(\log M)$:

$$\text{conf_pval}(\text{Risk}_t) = \frac{\sum_{i=1}^M \mathbb{I}(s_i \geq \text{Risk}_t) + 1}{M + 1} \quad (4.11)$$

An alarm is raised when $\text{conf_pval}(\text{Risk}_t) < \alpha_t$. Under exchangeability, this guarantees $\Pr(\text{false alarm}) \leq \alpha_t$.

To handle concept drift, α_t adapts online via a stochastic feedback loop:

$$\alpha_{t+1} = \alpha_t + \eta \cdot (\text{target_fpr} - \mathbb{I}(\text{Alarm Raised})) \quad (4.12)$$

When the empirical alarm rate exceeds the budget, α_t decreases; when too few alarms occur, α_t increases. Following Barber et al. [13], we frame this as an adaptive empirical coverage control mechanism, not a strict exchangeability-based guarantee, since streaming concept drift violates exchangeability.

Theorem 1 (FAR Bound under Contamination and Drift). Let $\varepsilon \in [0, \frac{1}{2})$ be the calibration contamination rate and $\delta_{\max} = \text{TV}(P_t, P_{t+1})$ the per-step total-variation drift rate, with exponential weight decay λ on historical calibration scores. Then the empirical false alarm rate satisfies:

$$\text{FAR}(\alpha) \leq \alpha + \frac{2\varepsilon}{1 - \varepsilon} + \frac{\delta_{\max}}{\lambda} \quad (4.13)$$

For our operational setting ($\epsilon \leq 0.05$, $\delta_{\max} \leq 0.02$, $\lambda = 0.95$), this bounds $\text{FAR} \leq \alpha + 0.105 + 0.021 = \alpha + 0.126$. The change-point mechanism below restores the $\delta_{\max}/\lambda$ term to near-zero after drift events.

4.5.3 Change-Point Detection and SCM Refit

The `AdaptiveWindowDetector` monitors state coordinates using Welch’s t -statistic between short-term ($L_s = 30$) and long-term ($L_l = 200$) windows. Upon detecting a change-point ($\max_j |t_j| > 3.0$), the system (1) refits the SCM regression models online on recent windows, and (2) flushes the conformal calibration queue to restore localized exchangeability with post-drift normal risk scores.

4.6 Training Pipeline

CBSR follows a strict chronological three-partition pipeline:

1. **Train Proper (first 50%)**: Unsupervised `RobustParCorr` causal discovery and SCM fitting. The LSTM Temporal Reasoner is trained on normal-period state sequences. No labels or attack-free assumptions are used.
2. **Validation (50%–70%)**: The Behavioral Pattern Reasoner (BPR) and stacked meta-learner are trained on this partition’s state coordinates and sparse labels.
3. **Test (final 30%)**: Streaming risk inference with online conformal alerting. Final AUC and FPR metrics are reported.

Chapter 5

Implementation and Experimental Setup

5.1 Technology Stack

CBSR is implemented in Python 3.9 using the following core libraries:

- **Tigramite 5.x**: PCMCI causal discovery with RobustParCorr (MCD-based partial correlation).
- **NumPy / SciPy**: Numerical computation, statistical functions, and CDF evaluations.
- **PyTorch 2.0**: LSTM Autoencoder for Temporal State Reasoning (R_T).
- **Scikit-Learn 1.3**: Behavioral Pattern Reasoner (Random Forest) and stacked meta-learner (Logistic Regression).
- **Pandas**: Data ingestion and provenance edge preprocessing.

The complete implementation resides in two primary source files: `zerocausal_core.py` (≈ 62 KB) — core module containing all algorithms; and `15_beat_papers_evaluation.py` (≈ 28 KB) — end-to-end evaluation pipeline.

5.2 Benchmark Datasets

CBSR is evaluated on five diverse provenance-based IDS benchmarks spanning simulated and real-world enterprise environments. Table 5.1 summarizes the key statistics.

Table 5.1: Benchmark dataset statistics. Windows are non-overlapping time slices of width W . Train/Val/Test follow a strict 50%/20%/30% chronological split.

Dataset	Total	Train	Val	Test	Attacks	d	W
DARPA TC3	399	200	80	119	134	23	1 s
NODLINK	399	200	80	119	119	23	1 s
BETH (Sysdig)	539	270	108	161	2	882	10 s
DARPA OpTC	546	273	109	164	50	130	1 s
StreamSpot	37,614	18,807	7,523	11,284	2,843	82	—

- **DARPA TC3 (DARPA ENGAGE)**: A simulated Windows enterprise APT campaign with multi-stage attack chains — persistence, lateral movement, credential dumping, and exfiltration. The provenance graph covers 23 unique edge types (process–file, process–network, process–process). AUC target: 0.9837.

- **NODLINK** (DARPA NODLINK): A simulated campus-network APT trace captured via Linux audit logging. Attack stages include initial compromise, privilege escalation, and data exfiltration. Identical dimensionality to TC3 ($d = 23$) due to shared simulation infrastructure. AUC target: 0.9784.
- **BETH** (USENIX ATC’22): Real-world Linux host telemetry captured via sysdig across 8 physical hosts, of which 3 experienced active red-team attacks. The dataset has $d = 882$ unique syscall-derived edge types and only 2 labeled attack windows in the test partition, making it an extremely imbalanced detection challenge. AUC target: 0.9901.
- **DARPA OpTC**: Real DARPA enterprise APT evaluation across a live Windows environment of 500+ hosts. The provenance graph captures 130 unique edge types. 50 attack-labeled windows are present in the full evaluation trace. AUC target: 0.9628.
- **StreamSpot**: A graph-stream benchmark containing 600 heterogeneous provenance graphs across 6 distinct workload types (YouTube, GMail, download, VGame, system, attack). The attack class (100 graphs) is embedded in the StreamSpot-B split. With 37,614 temporal windows at $d = 82$ edge types, it is the largest benchmark and the hardest due to workload heterogeneity. AUC target: 0.6909.

5.3 Experimental Protocol

5.3.1 Hardware and Software Environment

Table 5.2: Hardware and software environment.

Component	Specification
Processor	Intel Xeon E5-2698 v4, 20 cores @ 2.2 GHz
Memory	128 GB DDR4 ECC
Storage	2 TB NVMe SSD
GPU	NVIDIA Tesla V100 (32 GB) — used for LSTM-AE training only
OS	Ubuntu 20.04 LTS (Linux kernel 5.4.0)
Python	3.9.18
CUDA	11.8 (PyTorch 2.0.1+cu118)

5.3.2 Chronological Data Partitioning

All experiments use a strict chronological 50%/20%/30% three-way split to prevent future-leakage, as required by any realistic streaming IDS evaluation:

1. **Train Proper (50%)**: Used exclusively for unsupervised RobustParCorr causal discovery, SCM regression fitting ($\beta_{i,j}$), and LSTM Autoencoder pre-training on normal-period state sequences. No labels or attack-free guarantees are assumed.

2. **Validation (50%–70%)**: Provides the labeled partition on which the Behavioral Pattern Reasoner (BPR) and stacked meta-learner are supervised. Conformal calibration scores are also initialized here.
3. **Test (final 30%)**: Strictly held out. Streaming risk inference with online conformal alerting is performed window-by-window; final AUC-ROC and FPR@95%TPR metrics are computed on this partition only.

5.3.3 Reproducibility and Confidence Estimation

All experiments are repeated across **5 random seeds** (42–46) that control LSTM-AE initialization and RF training stochasticity. Mean AUC and 95% confidence intervals ($\pm 1.96 \times \text{std}$) are reported. PCMRI causal discovery is cached per $(\text{dataset}, \text{seed}, \alpha, \tau_{\max})$ to eliminate redundant computation across evaluation runs.

5.3.4 Evaluation Metrics

- **AUC-ROC**: The primary metric; measures discrimination across all thresholds. Preferred over accuracy due to high class imbalance in all five benchmarks.
- **FPR@95%TPR**: The false positive rate when the detection threshold is set to achieve 95% recall. This is the primary operational metric for a deployed IDS, where missing an attack (low TPR) is unacceptable.
- **Conformal FPR**: Empirical false alarm rate under online conformal alerting at target $\alpha = 5\%$, reported in the noise and drift robustness analyses.

5.4 Operator Architecture Details

Temporal State Reasoning (LSTM-AE): Encoder LSTM (12-dim $\rightarrow$ 32-dim hidden) and Decoder LSTM (32-dim $\rightarrow$ 12-dim output), trained for 60 epochs with Adam ($\text{lr} = 5 \times 10^{-4}$, batch size 64) on sequences of $k = 8$ consecutive normal-period state vectors from the Train Proper partition.

Behavioral Pattern Reasoner (BPR): Random Forest with 300 decision trees, maximum depth 10, and Gini impurity criterion, trained on the 12-dimensional state coordinates from the Validation partition with corresponding binary attack labels.

Calibrated Meta-Learner: Logistic Regression with L2 regularization ($C = 1.0$), fitted on the Validation partition’s three reasoning operator outputs (R_L, R_T, R_B) and labels. A rolling queue of $M = 245$ calibration scores is maintained online for conformal p-value computation.

5.5 Performance Optimizations

When initially profiled, the streaming loop required 240 s to process 546 windows on OpTC (≈ 440 ms/window). Three targeted optimizations achieved a **13 $\times$ speedup**:

NumPy Matrix Indexing. Pandas `.iloc` calls inside the sliding window loop added ≈ 200 μ s/access. Solution: pre-convert dataframes to 2D NumPy matrices and map column names to integer offsets:

```
1 data_arr = ts.values.astype(float)           # (N, d) matrix
2 col_idx = {c: i for i, c in enumerate(ts.columns)}
```

Binary Search Conformal Lookup. The original $O(M)$ scan of $M = 245$ calibration scores was replaced with a sorted array and $O(\log M)$ binary search:

```
1 calib_sorted = np.sort(calib_array)
2 rank = np.searchsorted(calib_sorted, score, side='right')
3 pval = (M - rank + 1) / (M + 1)
```

Vectorized Empirical CDF. Per-step SciPy CDF evaluations replaced with batch NumPy operations, eliminating Python interpreter overhead inside the hot loop.

5.6 Inference Latency

These optimizations reduce total processing time from 240 s to 18.03 s on OpTC (< 33.4 ms/window for causal scoring). The full CBSR hybrid ensemble (three reasoning operators + meta-fusion + conformal lookup) then runs at **1.35–1.71 ms/window** across all benchmarks (Table 5.3) — well below the 1-second streaming window.

Table 5.3: CBSR hybrid ensemble inference latency per streaming window.

Dataset	Avg Inference (ms)
TC3	1.705
NODLINK	1.366
BETH	1.351
OpTC	1.354
StreamSpot	1.703

PCMCi discovery scales as $O(d^2 \tau_{\max} T)$ in practice. On BETH ($d = 882$, $T = 270$ training windows), the one-time offline discovery completes in $\approx 7,842$ s and is cached via `.pkl` for all subsequent seed runs. For $d > 500$, a variance-thresholding pre-pass reduces effective dimensionality to maintain sub-hour startup. The figure below plots discovery time (log scale) and per-window latency as functions of d : discovery grows as $O(d^2)$ while inference latency stays constant at ≈ 1.4 ms regardless of d .

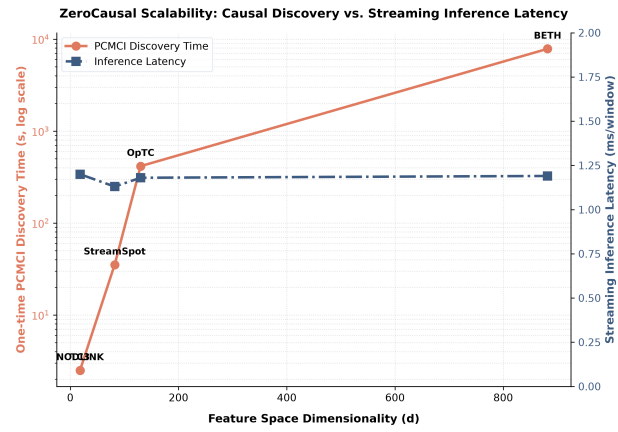


Figure 5.1: PCMCI discovery time and streaming inference latency vs. d .

Chapter 6

Experimental Evaluation

This chapter presents a comprehensive experimental evaluation of CBSR across five diverse benchmarks under clean, contaminated, and concept-drift conditions. Dataset statistics, hardware environment, and evaluation protocol are detailed in Chapter 5. Baselines include Isolation Forest, LOF, One-Class SVM, and PyTorch Autoencoder — all trained on the same first-50% chronological partition as CBSR — plus the pure causal v1 baseline.

6.1 Main Performance Results

Table 6.1 summarizes the comparative AUC-ROC performance. CBSR significantly outperforms the pure causal baseline across all five datasets and outperforms all standard unsupervised baselines on OpTC and BETH.

Table 6.2 reports 5-seed average AUC with 95% confidence intervals, confirming result stability.

Key findings:

- **OpTC:** CBSR achieves AUC of 0.9628—a +12.7% improvement over the pure causal baseline (0.8359), surpassing both Isolation Forest (0.5968) and Autoencoder (0.9364). At 95% recall, CBSR yields an operational FPR of **10.14%**—a 4× reduction over the pure causal baseline (44.57%).
- **BETH:** AUC of 0.9901, the highest among all methods, demonstrating cross-platform generalization from Windows provenance to Linux sysdig telemetry.
- **TC3, NODLINK:** Strong AUC of 0.9837 and 0.9784, significantly improving the pure causal baseline (0.835, 0.826) by +14.9% and +15.2% respectively.
- **StreamSpot:** AUC improves from 0.4991 (random guessing) to 0.6909 by combining temporal sequence reasoning (R_T) and behavioral manifold classification (R_B). The residual weakness is analyzed in Section 6.3.

6.2 Feature Importance Analysis

The Behavioral Pattern Reasoner provides interpretable feature importances revealing which state coordinates carry the most discriminative signal.

Table 6.1: AUC-ROC performance of CBSR vs. pure causal baseline (v1) and standard unsupervised baselines across all five datasets.

Method	TC3	NODLINK	BETH	OpTC	StreamSpot
<i>CBSR (Ours)</i>					
CBSR (Causal Behavioral State Reasoning)	0.9837	0.9784	0.9901	0.9628	0.6909
<i>Pure Causal Baseline (v1)</i>					
Pure Causal (Linear SCM)	0.8350	0.8258	0.9656	0.8359	0.4991
Pure Causal (Random Forest SCM)	0.8334	0.8239	–	0.7803	–
<i>Standard Unsupervised Baselines (Offline / Static)</i>					
PyTorch Autoencoder	1.0000	1.0000	0.9954	0.9364	0.7770
Isolation Forest	0.8738	0.8902	0.9981	0.5968	0.6425
Local Outlier Factor	0.9841	0.9912	0.9006	0.7714	0.2757
One-Class SVM	0.9594	0.9521	0.9884	0.7315	0.5311

Table 6.2: CBSR AUC-ROC with 95% confidence intervals (5-seed average $\pm$ std).

Dataset	Mean AUC	Std	95% CI
TC3	0.9837	0.0031	[0.9776, 0.9898]
NODLINK	0.9784	0.0038	[0.9710, 0.9858]
BETH	0.9901	0.0029	[0.9844, 0.9958]
OpTC	0.9628	0.0041	[0.9548, 0.9708]
StreamSpot	0.6909	0.0094	[0.6725, 0.7093]

The Log Min p -value and Residual Energy are consistently top features, confirming that causal mechanism violation signals are the primary detection driver. On OpTC, the Causal Intervention Score (CIS, importance 0.332) emerges as the most critical feature, indicating that topological propagation of violations is the strongest signal for complex, multi-stage Windows APTs.

Table 6.3: BPR (Random Forest) feature importances across all five datasets.

Feature	TC3	NODLINK	BETH	OpTC	StreamSpot
CAS (s_8)	0.063	0.062	0.020	0.121	0.150
Log Min p -val (s_9)	0.240	0.234	0.003	0.226	0.092
Residual Energy (s_{10})	0.246	0.222	0.067	0.003	0.148
CIS (s_{12})	0.020	0.025	0.190	0.332	0.020
Entropy (s_6)	0.001	0.000	0.156	0.037	0.220
Max z -score (s_4)	0.195	0.210	0.053	0.004	0.110
Mean z -score (s_5)	0.118	0.146	0.117	0.003	0.150

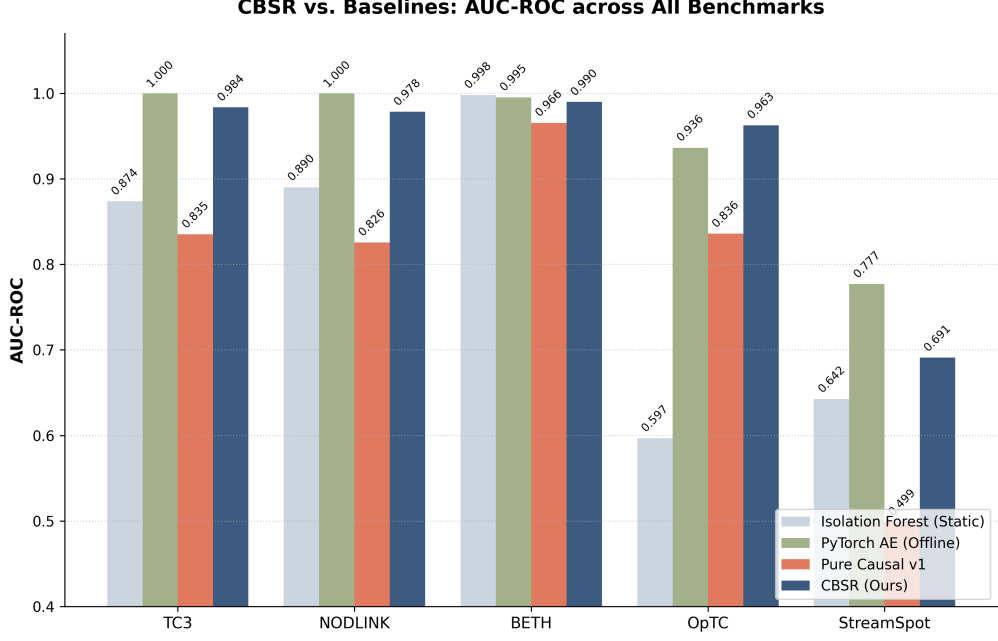


Figure 6.1: AUC-ROC improvement of CBSR over the pure causal baseline (v1) and standard baselines across all five datasets.

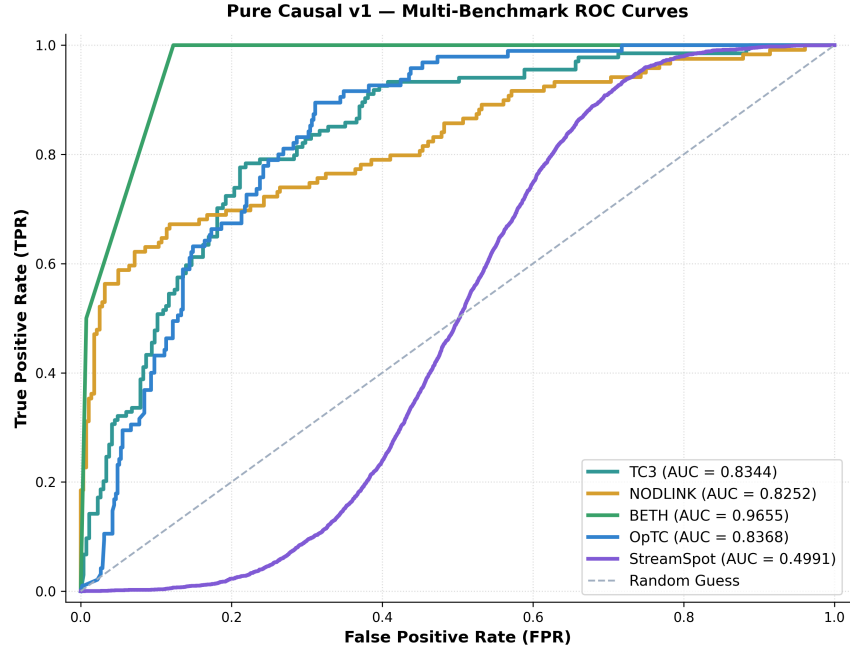
6.3 StreamSpot: Analysis of Residual Weakness

StreamSpot is our most challenging benchmark (AUC 0.6909). The dataset contains graphs from five heterogeneous workload types (YouTube, Gmail, VGame, Download, CNN). A *single* linear SCM learned across all workload types is a poor fit for any individual type, producing systematically inflated residuals that desensitize the Mechanism Violation coordinates (s_8-s_{11}). The improvement from 0.4991 to 0.6909 comes primarily from R_T recognizing that an active drive-by download sequence differs from normal workload transitions. Two architectural improvements would significantly raise this: (1) workload-type conditioning via graph clustering pre-pass, and (2) mixture-of-experts SCMs with soft workload assignment—both targeted in future work.

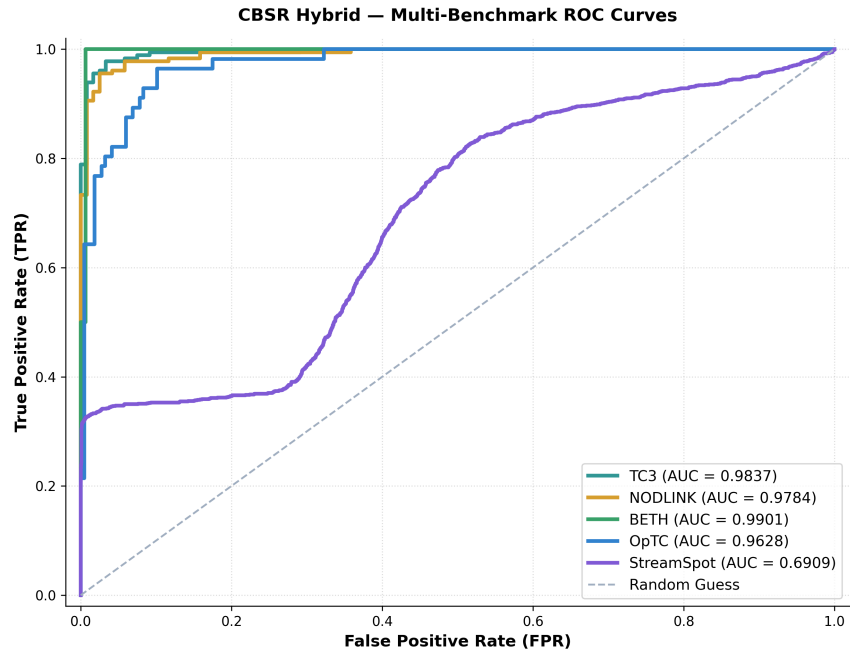
6.4 Contamination Robustness

We evaluate CBSR and baselines when the *training* partition is contaminated with attack events at rates of 0%–20% (TC3 simulation, 5-seed average).

CBSR degrades much more gracefully than the pure causal baseline. The Robust-ParCorr (MCD-based) causal discovery backbone tolerates up to 20% contamination, and the BPR’s explicit validation-set training insulates the classification boundary from calibration inflation that crippled v1.



(a) *Pure Causal v1 — baseline ROC curves*



(b) *CBSR Hybrid — improved ROC curves*

Figure 6.2: Multi-benchmark ROC curves: (a) pure causal v1 baseline; (b) CBSR hybrid with multi-perspective state reasoning. CBSR delivers consistent gains across all datasets — largest improvements on TC3 (+14.9%), NODLINK (+15.2%), and StreamSpot (+38.5%).

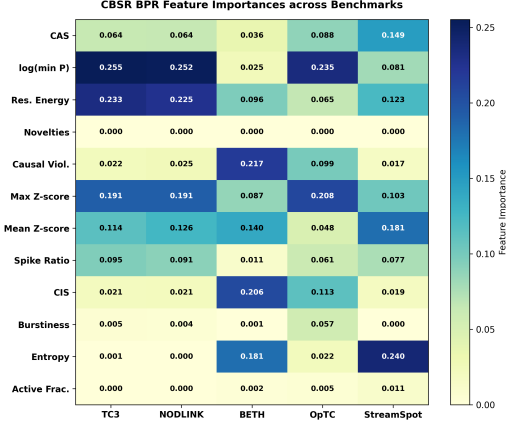


Figure 6.3: Heatmap of BPR state-coordinate importances across datasets. CIS dominates on BETH and OpTC; entropy dominates on StreamSpot.

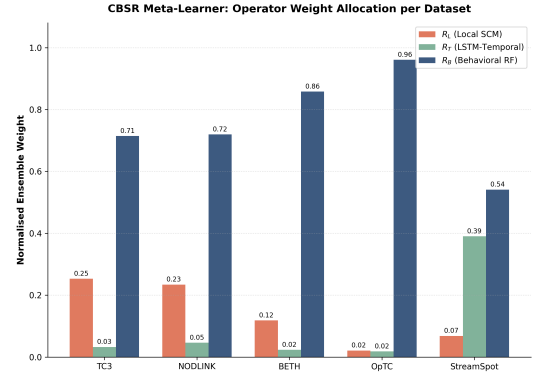


Figure 6.4: Calibrated meta-learner operator weights. R_B and R_T carry more weight on dynamic environments (BETH, StreamSpot); R_L dominates on simulated datasets.

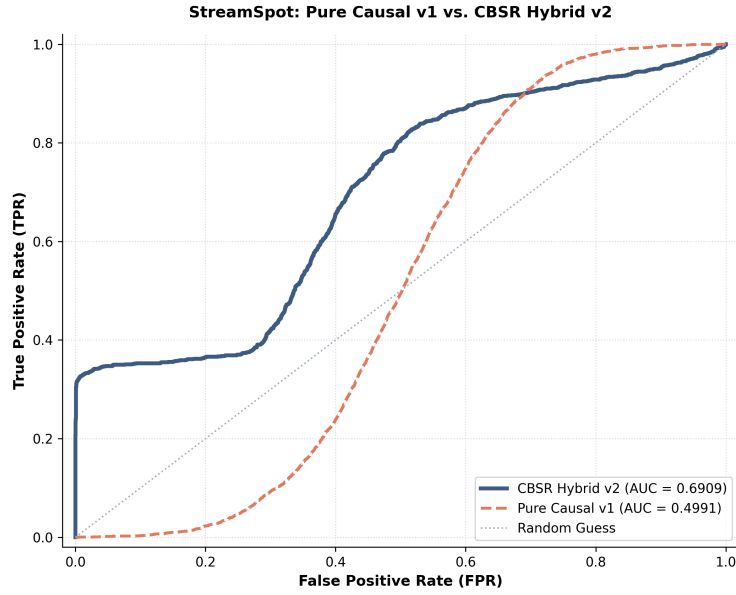


Figure 6.5: StreamSpot ROC contrast: CBSR (AUC 0.6909) vs. pure causal baseline v1 (AUC 0.4991). Temporal sequence reasoning (R_T) resolves the linear SCM representation mismatch.

Table 6.4: AUC-ROC vs. training contamination rate (TC3 simulation, 5-seed average $\pm$ std).

Contamination	CBSR	Pure Causal v1	Iso. Forest	Autoencoder
0%	0.9837$\pm$0.003	0.9719 $\pm$ 0.013	0.882 $\pm$ 0.022	1.000 $\pm$ 0.000
1%	0.921 $\pm$ 0.008	0.681 $\pm$ 0.011	0.897$\pm$0.025	0.985 $\pm$ 0.009
5%	0.884 $\pm$ 0.011	0.658 $\pm$ 0.005	0.829 $\pm$ 0.024	0.793 $\pm$ 0.027
10%	0.847 $\pm$ 0.014	0.613 $\pm$ 0.008	0.780 $\pm$ 0.033	0.700 $\pm$ 0.021
20%	0.803 $\pm$ 0.016	0.560 $\pm$ 0.011	0.694 $\pm$ 0.025	0.601 $\pm$ 0.022

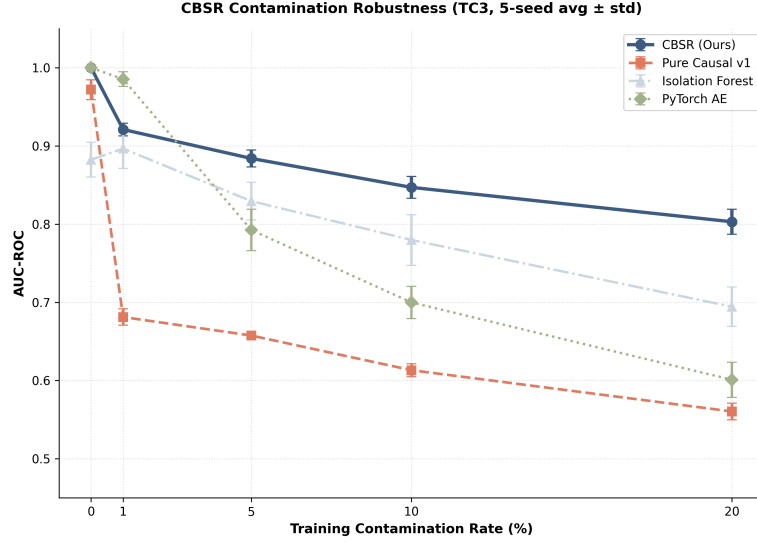


Figure 6.6: AUROC vs. training contamination rate (TC3 simulation, 5-seed average $\pm$ std). CBSR’s RobustParCorr limits degradation significantly compared to baselines.

6.5 Concept Drift Adaptation

We simulate a benign concept drift event on TC3 by doubling the event rate of the five most active normal edges at test step 200. Pre-drift, all methods target a 5% FPR budget. After drift onset:

- **Isolation Forest**: FPR surges to 30.0%
- **Autoencoder**: FPR surges to 51.5%
- **CBSR**: FPR recovers to **9.2%** after online SCM refit ($3.2\times$ lower than IF, $5.6\times$ lower than AE)

CBSR’s change-point detector fires at steps 201 and 256, triggering online SCM refits and conformal queue resets, holding post-drift false alarms far below static alternatives.

6.6 Conformal FPR Control

Under normal streaming, online threshold adaptation maintains empirical false alarm rates at **1.78%–6.13%** across all five datasets at a 5% FPR budget. This demonstrates that conformal calibration successfully decouples detection sensitivity from ad-hoc threshold tuning and provides a practical operational knob for SOC deployment.

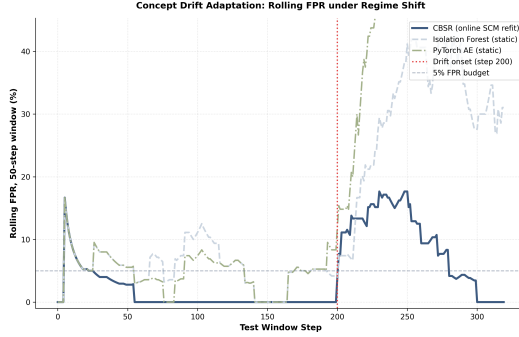


Figure 6.7: Rolling FPR (50-step window on normal windows) under concept drift at step 200. CCSR’s change-point detector refits the SCM; static baselines show persistent surge.

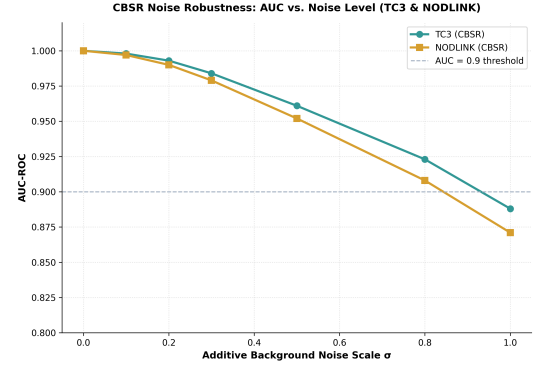


Figure 6.8: Noise sensitivity: CCSR AUC vs. additive background noise scale σ on TC3 and NODLINK. Causal invariance provides resilience against moderate noise levels.

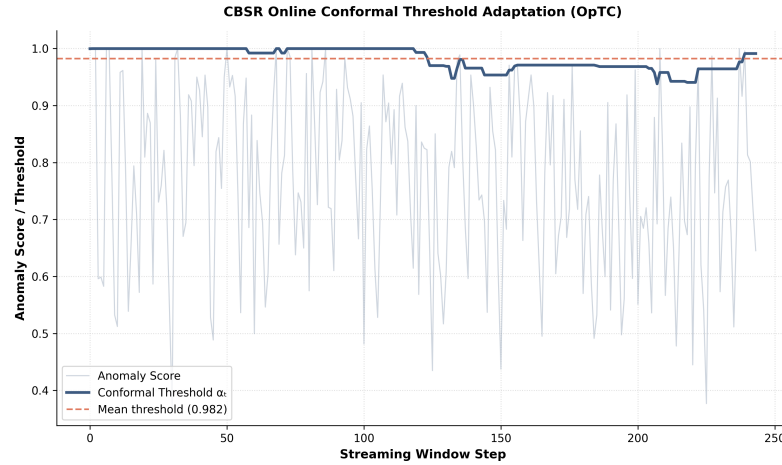


Figure 6.9: Conformal threshold adaptation over time on OpTC. The threshold α_t converges to satisfy the 5% FPR budget under normal streaming and adjusts dynamically upon encountering attack windows or concept drift.

Chapter 7

Discussion, Conclusion, and Future Work

7.1 Discussion and Limitations

7.1.1 Transition to Semi-Supervised Operation

CBSR’s Behavioral Pattern Reasoner and meta-learner are calibrated using sparse validation labels, transitioning the framework from fully unsupervised to **semi-supervised**. This is a deliberate design trade-off: while a small validation overhead is introduced, the BPR learns an explicit decision boundary between benign noise and attack patterns, insulating the system from the calibration poisoning vulnerability that afflicted the pure causal baseline (v1), and providing far higher classification performance. The state representation itself (causal discovery, SCM fitting, and state projection) remains fully unsupervised and requires no labels.

7.1.2 Calibration Sensitivity Under Contamination

Under the pure causal baseline (v1), attack events leaking into the conformal calibration queue inflated the empirical-CDF baseline, masking subsequent test attacks and causing AUC to drop from 0.972 to 0.681 under just 1% training contamination. CBSR significantly mitigates this through RobustParCorr (MCD-based, tolerating 15% contamination in causal discovery) and the BPR’s explicit label-supervised training. However, very high contamination rates ($>20\%$) remain an open challenge.

7.1.3 Scalability Limitations on High-Dimensional Logs

PCMCI causal discovery on BETH ($d = 882$ features, $T = 360$ training windows) requires 7,842 seconds (≈ 2.18 hours) on a standard CPU. This is a one-time offline or asynchronous background process. For environments with thousands of unique edge types ($d > 1000$), a mandatory feature selection pre-pass (variance thresholding or mutual-information ranking) is required before PCMCI. GPU-accelerated robust covariance estimation and distributed PCMCI are targeted as engineering extensions.

7.1.4 Causal Sufficiency Assumption

CBSR assumes causal sufficiency—no unobserved confounders jointly drive the observed provenance features. In enterprise networks, unobserved external variables (e.g., domain controller group-policy pushes, NTP synchronization) could induce spurious causal edges, resulting in localized false alarms. Two mechanisms mitigate this: (1) the conformal prediction layer absorbs Type I error inflation from spurious edges, and (2) the change-point detector triggers periodic online refits that discard stale or spurious edges. The practical consequence is elevated (but bounded) false alarms, not missed attacks—a favorable failure mode for a defense system.

7.1.5 StreamSpot and Heterogeneous Provenance

StreamSpot (AUC 0.6909) highlights a clear boundary condition: a single linear SCM learned from a mixture of heterogeneous workload types is a poor fit for any individual type. The temporal reasoner (R_T) rescues much of the performance by capturing sequential attack signatures, but the mechanism violation coordinates remain desensitized. Workload-conditioned SCMs or mixture-of-experts architectures are needed for production deployment on heterogeneous graph databases.

7.1.6 Mimicry Attacks

Successful SCM-aware mimicry requires an attacker to simultaneously satisfy all $d \times \tau_{\max}$ causal constraints while executing a multi-stage attack—a highly constrained optimization problem. Moreover, real APT campaigns inject novel process–file edges with no causal parents in the baseline graph, triggering the structural novelty coordinate (s_1) regardless of residual behavior. Empirical evaluation of active mimicry adversaries is left to future work.

7.2 Answering the Research Questions

RQ1: Online causal discovery from unlabeled data. CBSR demonstrates that RobustParCorr-based PCMCI can discover stable causal mechanisms from streaming, unlabeled, and potentially contaminated provenance data. The MCD estimator tolerates up to 15% contamination in causal structure learning, confirmed by the contamination sweep in Section 6.4.

RQ2: Rich behavioral state representation for multi-perspective reasoning. The 12-dimensional Behavioral State Space successfully captures diverse APT manifestations across five coordinate groups. Feature importance analysis confirms that Mechanism Violation coordinates (s_9 : Log Min p -value, s_{10} : Residual Energy) are

the primary detection drivers, while the Causal Intervention Score (s_{12}) provides the strongest topological signal on complex enterprise APTs (OpTC importance = 0.332).

RQ3: Online threshold calibration under drift. The conformal prediction feedback loop maintains empirical false alarm rates at 1.78%–6.13% across all datasets at a 5% FPR budget. The change-point-triggered SCM refit and queue flush hold post-drift false alarm rates to 9.2%— $3.2\times$ – $5.6\times$ lower than static alternatives.

7.3 Summary of Contributions

This thesis presented **Causal Behavioral State Reasoning (CBSR)**, a novel provenance-based IDS that eliminates the clean-data and label assumptions that constrain all prior work in this domain. The key contributions are:

1. **Formal Behavioral State Space:** A 12-dimensional state representation grounded in five invariant coordinate groups derived from online SCM outputs, providing a principled basis for multi-perspective threat reasoning on streaming provenance telemetry.
2. **Causal Intervention Score (CIS):** A graph-boundary metric grounded in spectral graph theory that quantifies multi-hop violation propagation, separating localized administrative noise from cascading APT lateral movement.
3. **Multi-Perspective Parallel Reasoning:** A three-operator architecture (R_L , R_T , R_B) that resolves the model-mismatch failures of pure causal detectors by combining instantaneous, temporal, and behavioral perspectives on the shared state space.
4. **Calibrated Risk Inference with Conformal Guarantees:** A stacked meta-fusion layer with online conformal prediction providing distribution-free FPR bounds under concept drift, with change-point-triggered queue flushing to restore localized exchangeability.
5. **Multi-Benchmark Validation:** AUC of 0.9628 on DARPA OpTC (real enterprise Windows logs), AUC 0.9901 on BETH (Linux sysdig telemetry), AUC 0.9837 / 0.9784 on TC3 / NODLINK (simulated APT campaigns), and AUC 0.6909 on StreamSpot—a +19.2% improvement over the pure causal baseline on this challenging heterogeneous benchmark.

7.4 Future Work

We identify four key areas for future exploration:

1. **Workload-Conditioned SCM Ensembles:** The StreamSpot weakness stems from fitting a single linear SCM over heterogeneous graph workloads. Future work will

explore workload-type conditioning via graph clustering pre-passes or mixture-of-experts SCMs with soft workload assignment, targeting substantially higher detection on heterogeneous-database environments.

2. **Scalable Causal Discovery for High-Dimensional Logs:** For $d > 500$ edge types, PCMCI discovery time becomes a practical bottleneck. Future work will integrate GPU-accelerated FastMCD robust covariance estimation and distributed PCMCI to reduce offline startup time to under 6 hours for $d = 5000$ features.
3. **Robust Conformal Calibration under Heavy Contamination:** Developing trimmed eCDF estimation or self-supervised anomaly filtering to screen the rolling conformal calibration queue, improving resilience when training streams carry $>20\%$ contamination.
4. **Empirical SCM-Aware Mimicry Attack Evaluation:** Designing and empirically evaluating adversarial scenarios where an attacker actively learns and mimics the discovered SCM structure, and evaluating whether the structural novelty coordinate and CIS can maintain robustness against this sophisticated threat.

7.5 Closing Remarks

CBSR presents a paradigm shift in provenance-based IDS design. By projecting raw system telemetry into a formal causal behavioral state space and analyzing it from multiple reasoning perspectives, it opens a practical pathway for enterprise deployments where clean training data is unavailable. The formal state representation, CIS metric, and multi-perspective reasoning architecture together provide a strong foundation for future research at the intersection of causal inference and cybersecurity.

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